

Basics of Network

Network Criteria: Performance, Reliability, Security

Type of Connection: Point-to-point, Multipoint

Physical Topology: Mesh, Star, Bus, Ring

Lecture # 3 and 4

Week# 2

Data Communications and Networking, Fourth Edition, Behrouz A. Forouzan

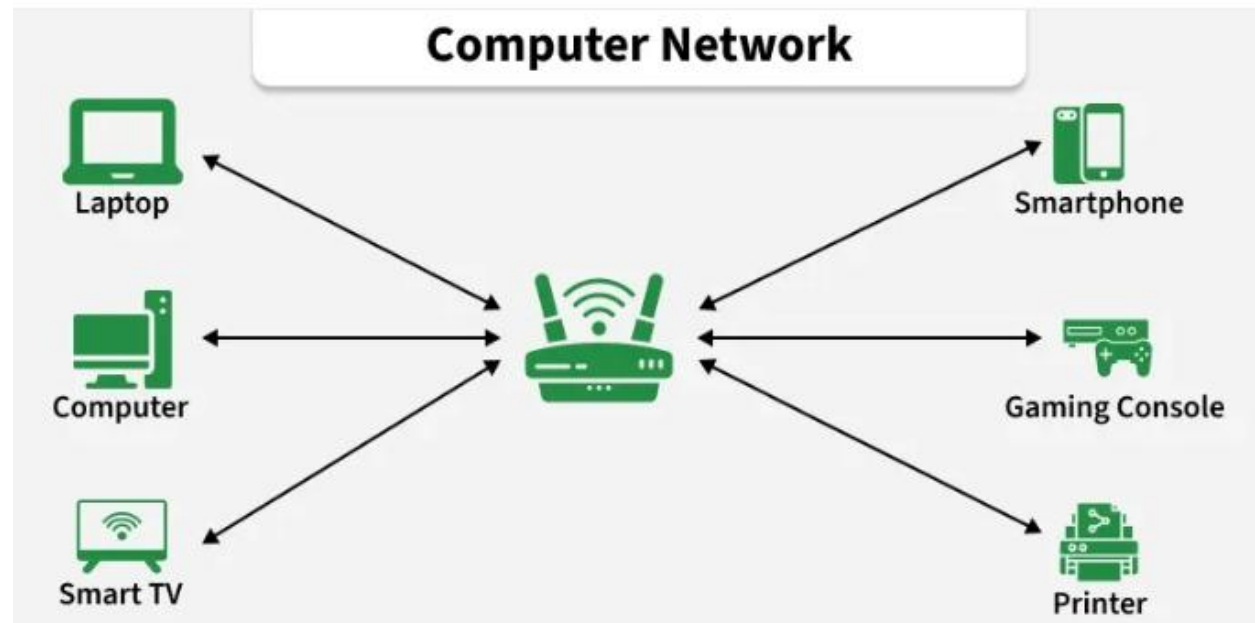
Prepared by:

Mehak Rana

Lecturer, Department of Computer Science

The University of Faisalabad

- A computer network is a system of interconnected devices, such as computers, servers, smartphones, and printers, that communicate to exchange data. They communicate using wired or wireless connections and can range from small home networks to the global Internet
- **Simple Example:**
- Computers in a school lab connected to one printer
- Mobile phones connected to the same Wi-Fi
- ATMs connected to bank servers
- All these are **computer networks**

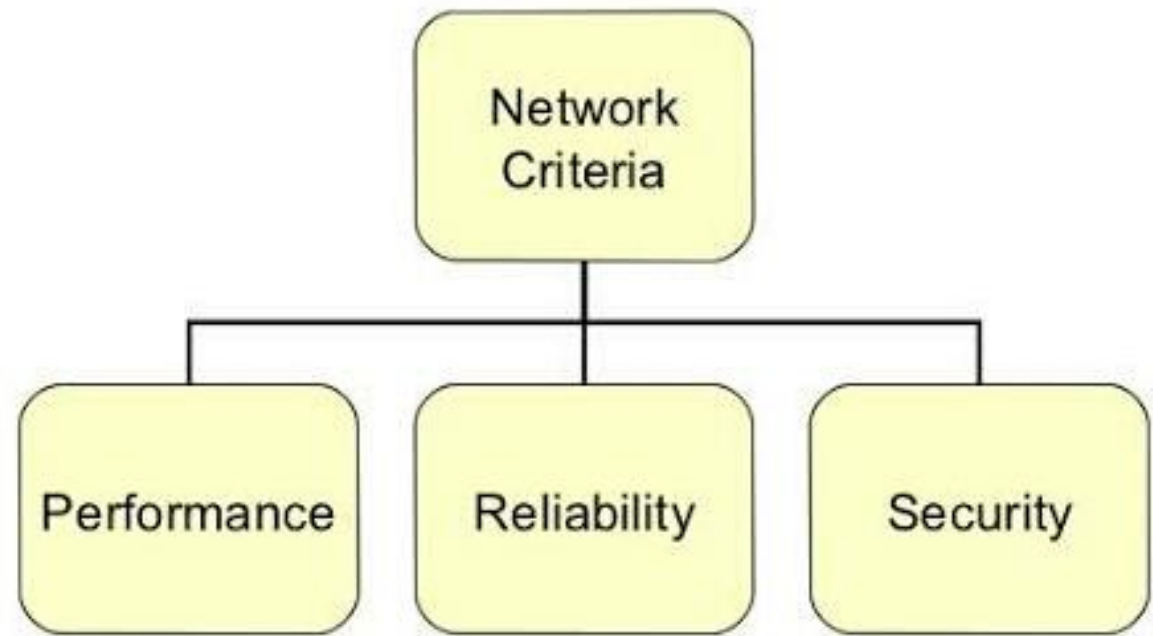


- These are standards used to judge how good a network is

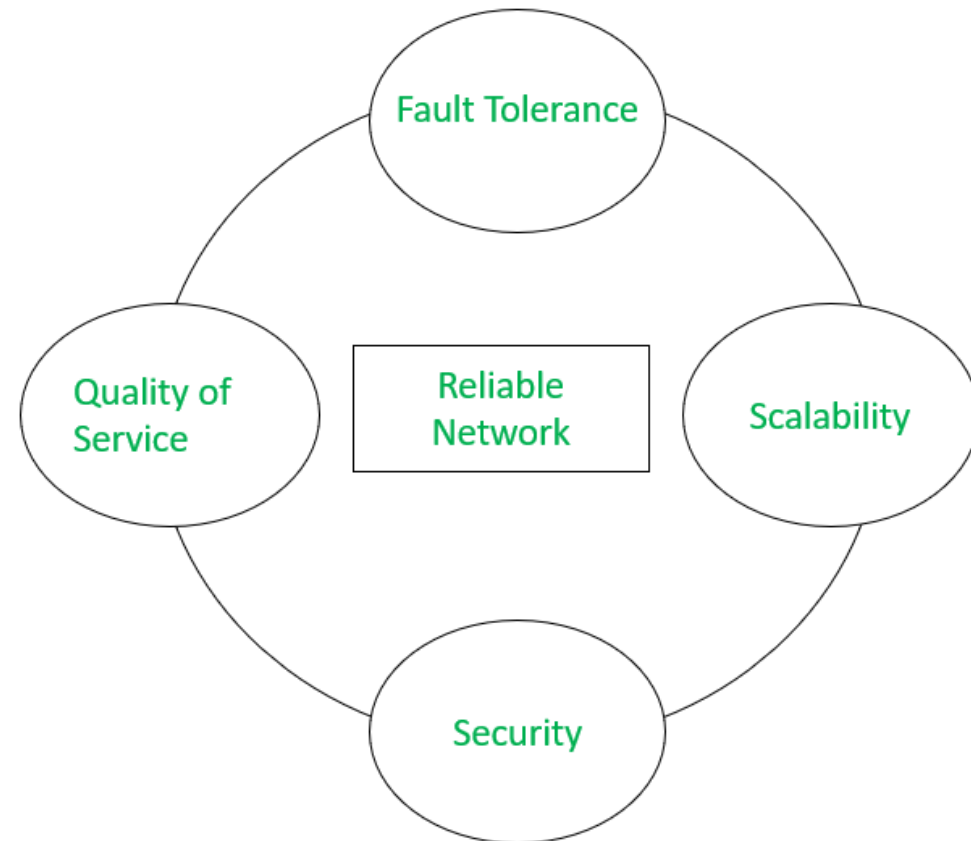
Performance

- **Meaning:** How fast data is sent and received.
- Includes:
 - Speed (bandwidth)
 - Delay (latency)
 - Response time
- **Real-life example:**

If your online class video loads smoothly without buffering, the network performance is good.



- **Meaning:** How dependable the network is (how often it fails).
- Includes:
 - Frequency of failure
 - Recovery time
 - Backup links
- **Real-life example:**
 - University Wi-Fi working all day without disconnecting = reliable network.



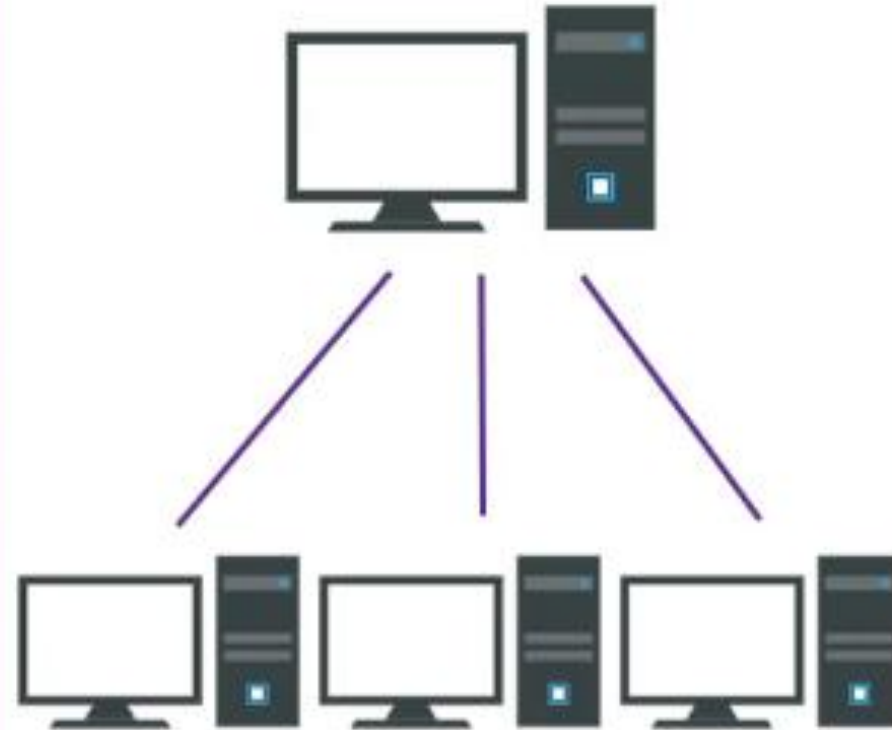
- **Meaning:** Protecting data from unauthorized access.
- Includes:
 - Passwords
 - Firewalls
 - Encryption
- **Real-life example:**
Only students with login credentials can access LMS portal → security.

Type of Connection

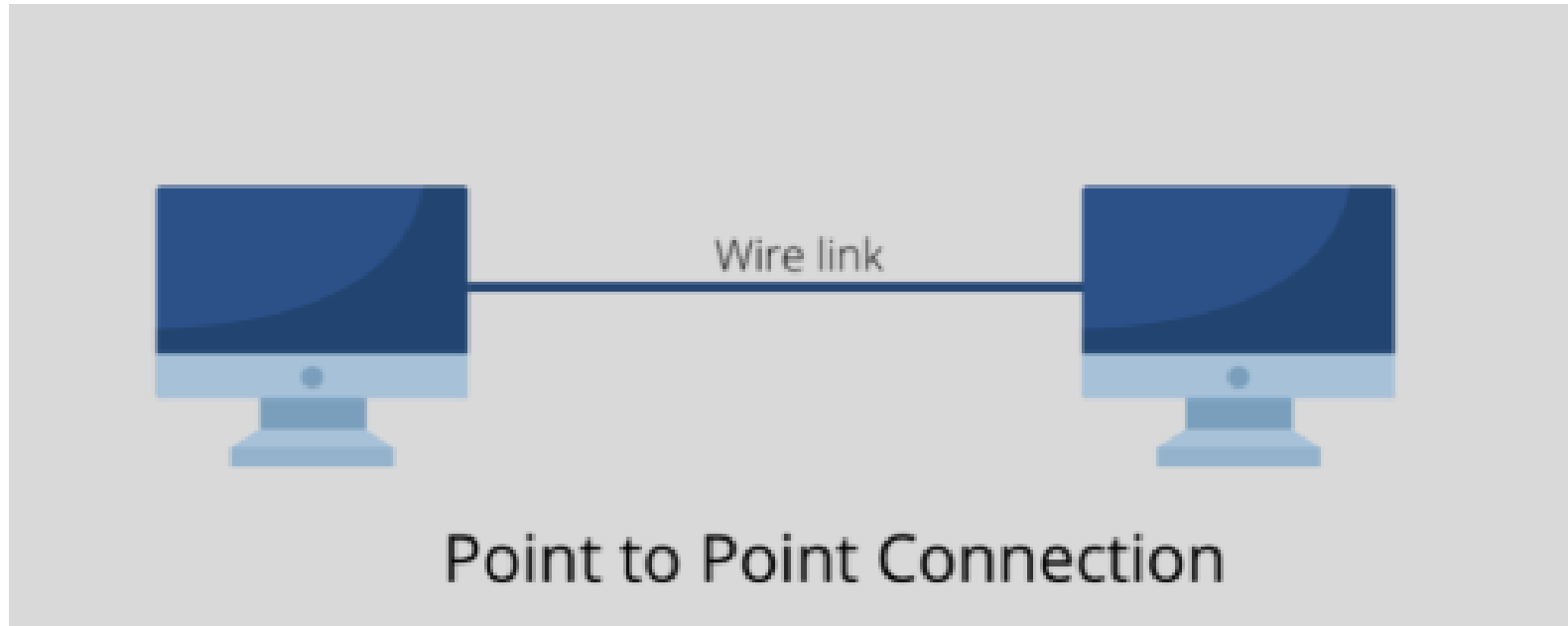
Point to Point



Point to Multipoint

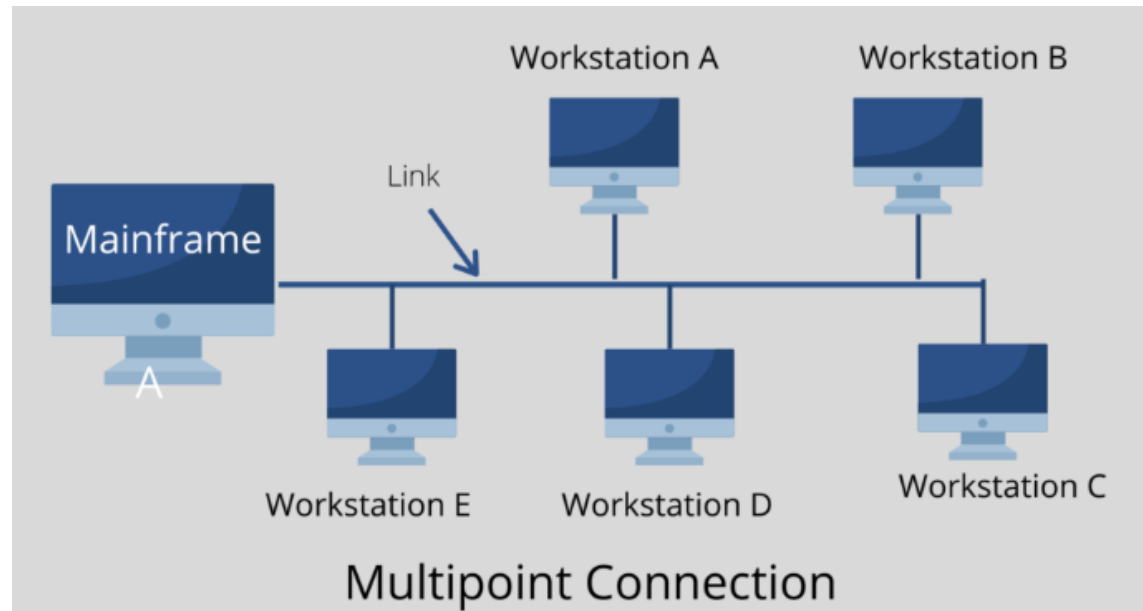


- Point-to-Point
- **Meaning:** Direct connection between two devices.
- Only two devices communicate
- **Example:**
Computer connected directly to printer .



Multipoint

- A **Multipoint** connection is when **more than two devices share the same communication link**.
- Many devices communicate together.
- **Example:**
- Many students using one Wi-Fi router
- Computers in a lab connected to one switch
- Multiple mobiles connected to same hotspot

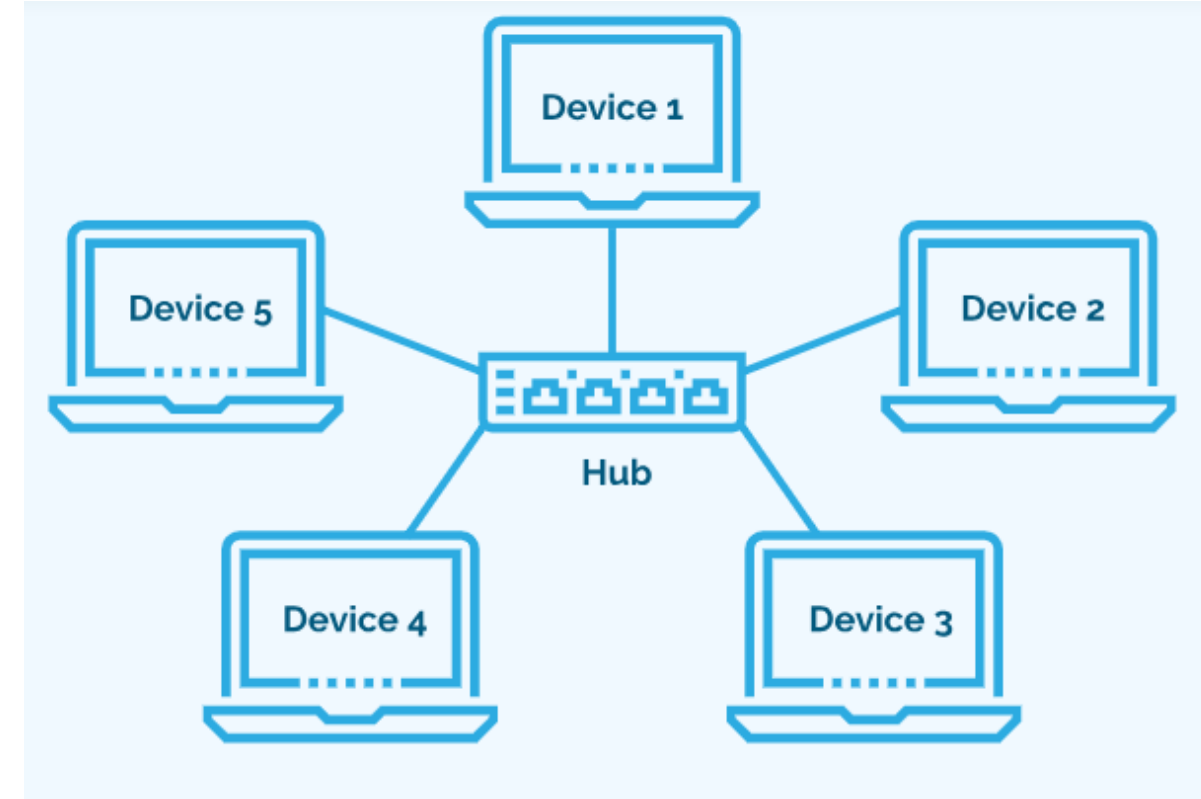


- **Network topology** is the **physical or logical arrangement (layout)** of computers and cables in a network.
- There are mainly eight types of topology in computer networks:
 1. Mesh Topology
 2. Star Topology
 3. Bus Topology
 4. Ring Topology
 5. Hybrid Topology
 6. P2P Topology
 7. Daisy Chain Topology

Star Topology

Definition

- All devices connect to a **central hub or switch**.
- **Real-life Example**
- Computer labs in schools.
- **Advantages**
- Easy to install and manage
- One cable failure doesn't affect others
- Easy to add new computers
- **Disadvantages**
- If hub fails → entire network stops
- Needs more cable



Definition

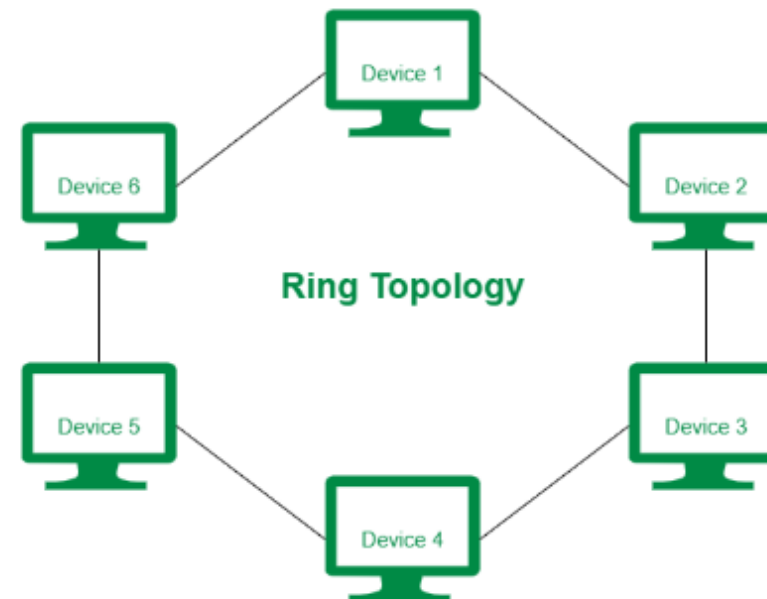
- Devices are connected in a **circular form** and data moves in one direction.
- **Real-life Example**
- In some cities, the internet company connects buildings in a circle using fiber-optic cable, so the signal goes around the loop until it reaches your home.

Advantages

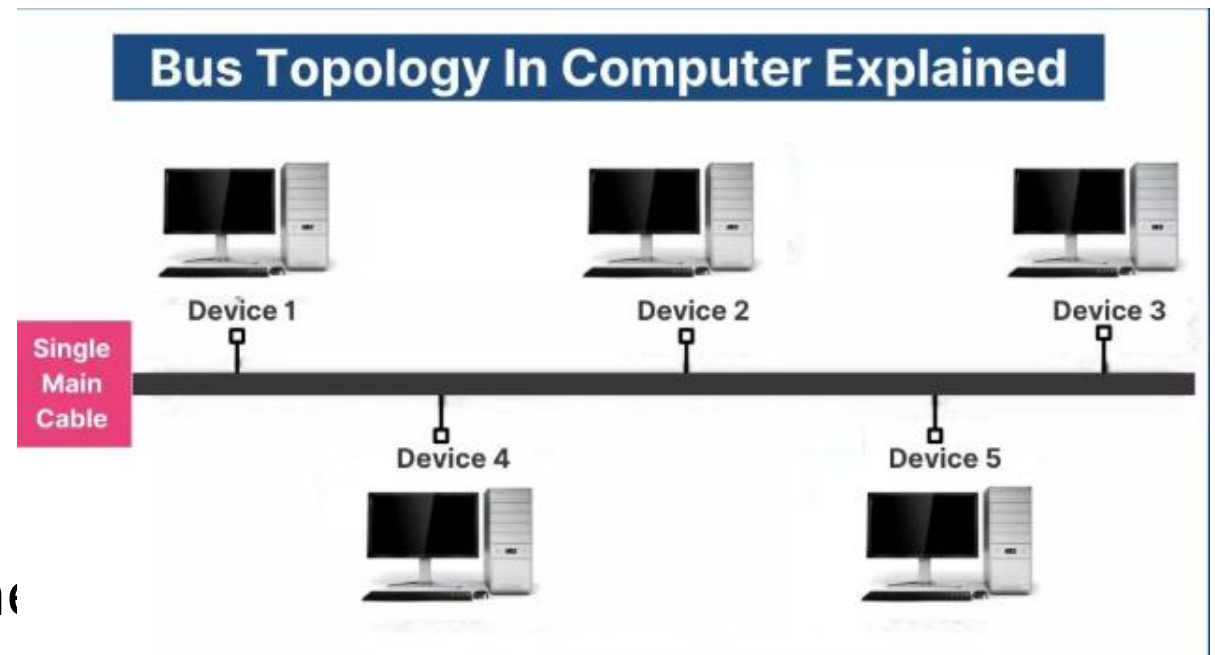
- Equal access for all devices
- No data collision

Disadvantages

- One device failure affects whole network

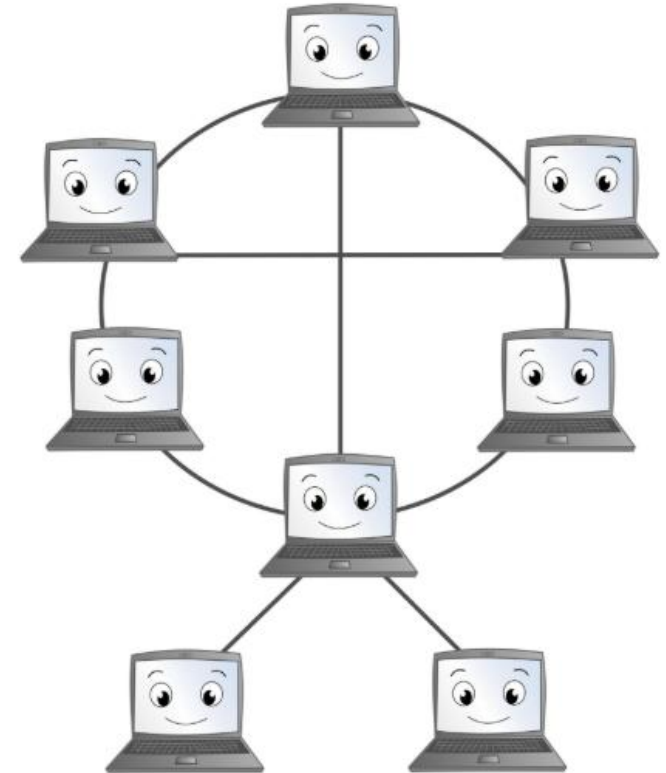


- **Definition**
- All devices share a **single main cable (backbone)**.
- **Real-life Example**
- A cable TV line runs along the street, and every house connects to the same main wire. If the main wire is cut, everyone loses the signal
- **Advantages**
- Cheap
- Simple for small networks
- Uses less cable
- **Disadvantages**
- If main cable breaks → network fails
- Limited number of devices can be connected
- Low security



Mash Topology

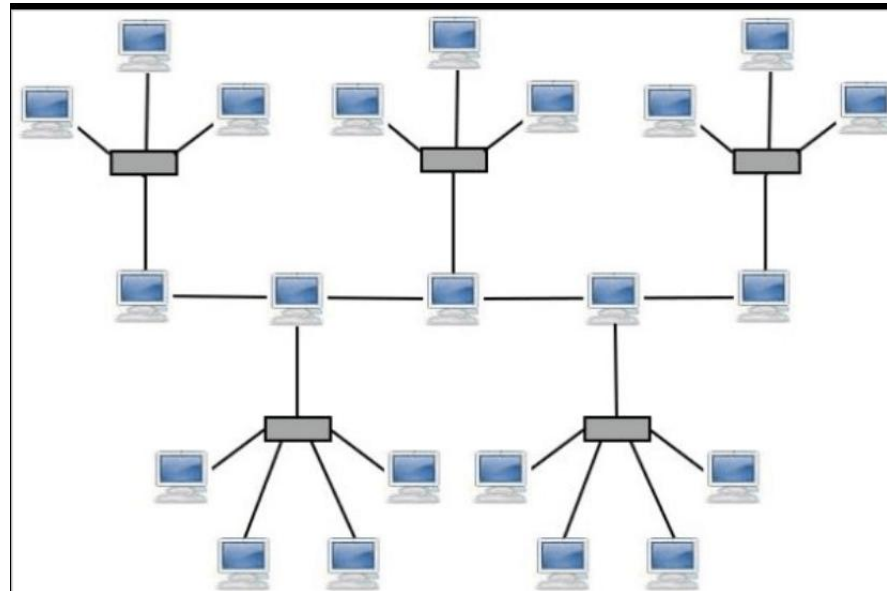
- In **Mesh topology**, every device is connected to **many or all other devices**.
- **Example:**
- Imagine a bank ATM machine is connected to **multiple bank servers/branches**:
- ATM $\leftrightarrow$ Branch A
- ATM $\leftrightarrow$ Branch B
- ATM $\leftrightarrow$ Main Bank Server
- Now:
- If Branch A is unavailable or busy
→ The ATM automatically sends the request to Branch B or the main server
→ You can still withdraw money
- **Advantages**
- Very reliable (no single point of failure)
Fast communication
Data can take alternate path
- **Disadvantages**
- Very expensive
Complex setup



- **Definition**
- Hybrid topology is a **combination of two or more topologies**(Star, Bus, Ring, Mesh).
- **Real-life Example**

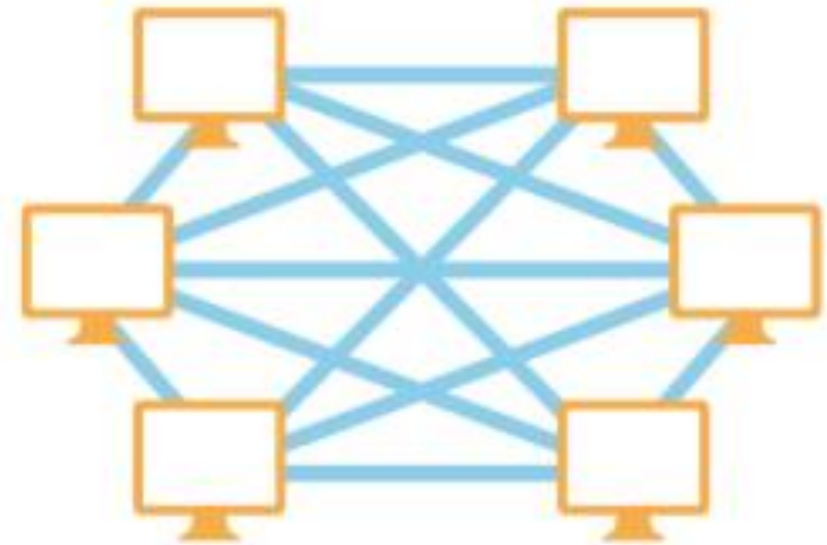
In a university network, the computers inside each faculty Rooms are connected in a star topology using a switch. Then all Faculty Rooms are connected to one main cable using bus topology. Together, this forms a hybrid topology.

- **Advantages**
- Flexible design
- Reliable
- Scalable
- **Disadvantages**
- Expensive
- Complex setup



- **P2P (Peer-to-Peer) Topology**
- **Definition**
- All computers are **equal** and share resources directly without central server.
- **Real-life Example**
- **Two Laptops Sharing Files**
- You connect two laptops using a LAN cable or Wi-Fi hotspot.
- Laptop A sends movie to Laptop B
- No router server needed
- Both computers directly talk to each other
- **Advantages**
- Easy setup
- No server needed
- Low cost
- **Disadvantages**
- Less secure
- Not suitable for large networks

Peer To Peer Network (P2P)



- **Daisy Chain Topology**
- **Definition**
- Devices are connected **one after another in a line.**
- **Example**
- Printers or monitors connected in series.
- **Advantages**
- Simple setup
- Uses less cable
- **Disadvantages**
- If device fails, others stop
- Limited expansion



THANK YOU

You can Find me:

mehakrana.CS@tuf.edu.pk